

ASSISTIVE ROBOTICS TEAM 1 REPORT

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ABSTRACT

This project report addresses the challenge of hearing loss, affecting a significant global population, by developing an assistive audio-to-touch conversion device. This technology is crucial as a large population suffers from hearing loss whilst age and occupational noise are major factors in hearing loss. The device employs stepper motors and solenoids within transmitting pads, each tuned to different frequency ranges (low, mid, high), enabling the detailed translation of music into tactile sensations for those with hearing impairments. Our methodology involved audio file processing to isolate harmonic and percussive elements, utilizing Fast Fourier Transform (FFT) and Short Time Fourier Transform (STFT). The device's operational logic aligns frequency intervals with corresponding stepper motors and solenoids to create a tactile music experience. The research demonstrates the potential of this technology in enhancing musical enjoyment for the hearing-impaired, though further refinements in design and functionality are needed for improved user experience and reliability. This work tries to contribute to the field of assistive technologies, offering a novel approach to make music more accessible and enjoyable for individuals with deafness in all spectrums in general.

INTRODUCTION

Hearing loss is a prevalent issue, affecting a significant portion of the global population. In the United States, it is estimated that 2 to 3 out of every 1,000 children are born with detectable levels of hearing loss in one or both ears. Approximately 15% of American adults also report some hearing difficulties[1]. Notably, over 90% of deaf children are born to hearing parents, which highlights the need for innovative communication aids. Age is a major factor in hearing loss, particularly among those aged 60 to 69, with men being more affected than women. Occupational noise exposure is another significant cause, with about 18% of adults aged 20-69 experiencing speech-frequency hearing loss after prolonged exposure to loud environments. Alarmingly, around one in eight people in the U.S. aged 12 and older have hearing loss in both ears, based on standard hearing examinations[2]. This situation emphasizes the necessity for new solutions to help those with hearing impairments, especially in experiencing music and sound more fully.

The development of assistive technologies for individuals with hearing impairments showcases significant progress. The initial focus was on basic tactile audio displays that translated audio signals into vibration patterns, allowing individuals to perceive sound through touch. Technology has since evolved, leading to more sophisticated systems like the EmotiChair project, which translates music into complex vibration patterns. This evolution indicates a growing understanding of the needs of the hearing-impaired community and the potential for further innovations in this area.

In haptic interface development, stability, high haptic quality, and usability are crucial. These systems must provide stable and reliable feedback that is meaningful and satisfying for the user. The challenge lies in balancing technical goals with the sensory capabilities of the users. Usability is vital, as the systems must be efficient and satisfying to use. Recent methodologies in haptic system development include comprehensive data management for simulation models, indicating the increasing complexity of this field.

Recent studies in haptic feedback for music experience demonstrate technology's role in enhancing musical enjoyment, particularly for those with hearing impairments [3]. These advancements focus on using haptic technology to translate music's rhythm, tempo, and intensity into tactile sensations, allowing deaf individuals to 'feel' the music. The development of customizable haptic devices caters to the varied spectrum of deafness, ensuring a personalized music experience. The integration of haptic feedback with visual elements also enriches the experience for deaf individuals, making music more inclusive.

The current project objective is to develop a device that enables deaf individuals to experience the full spectrum of musical frequencies, not just bass vibrations commonly felt in loud environments. This device integrates stepper motors and solenoids within three transmitting pads, each tuned to different frequency ranges: low, mid, and high. This design allows for a detailed translation of music into tactile sensations. The focus extends beyond technological innovation to emphasize user accessibility and comfort. The goal is to create a user-friendly device that translates musical frequencies into tactile vibrations, opening up new musical appreciation for those with hearing impairments and making music accessible to all.

METHODOLOGY

Hardware Design

When building the hardware, it was critical to enable the ability to emulate tones and percussions, to isolate vibrations from one actuator to another, and to match the ergonomic shape of a hand while keeping all parts within a \$200 budget.

To give people with profound hearing loss the ability to feel music, a set of stepper motors and solenoids were used to emulate tones and percussions respectively. There were three zones, where each zone had a stepper motor and a solenoid. Each zone would vibrate for a different range of frequencies including bass, mid, and treble. This way, a song in an audio file format could be converted to a series of instructions for the stepper motors and solenoids to vibrate.

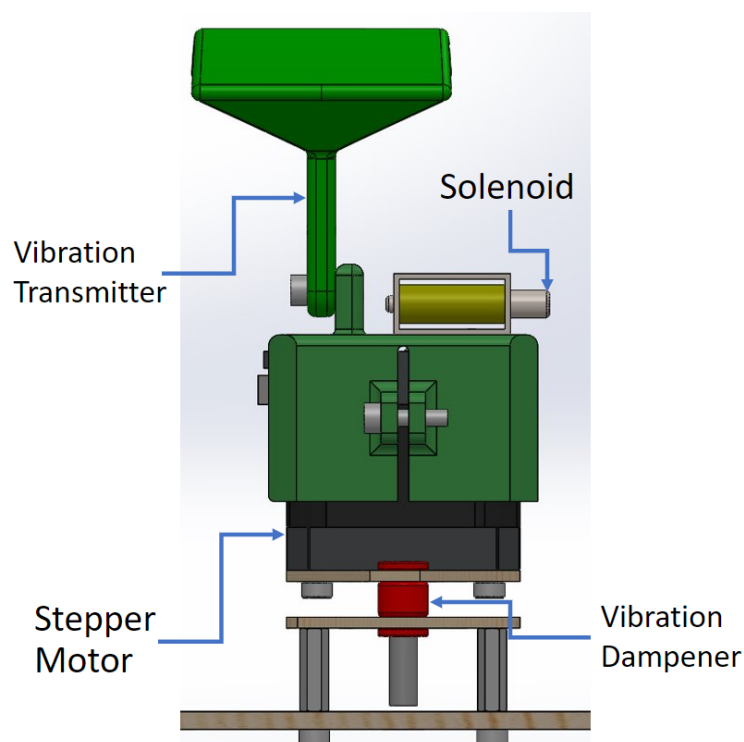


Figure 1: CAD model of Isolated Vibration unit

Each zone, alongside the actuators, included several parts to make the assembly stable, to isolate vibrations, to guide the vibrations to the user's hand, and to contain all actuators within a small volume. The stepper motor worked as a centre of the zone assembly. Since the stepper motor had an elongated shaft, the stepper was mounted offset from the device's frame with standoffs. Then, the stepper motor was isolated further with high durometer silicone balls. These balls acted as vibration dampeners and reduced the effects of one motor interfacing with another. Then to transmit vibrations from the stepper motor to the hand, a 3D-printed clamp was designed. The clamp wrapped around the motor and included two screw holes to hold the solenoid in place. Finally, a nub was added on top of the clamp to hold a vibration transmitter. This vibration transmitter included a long thin section and a large area, like a pad, to transmit the vibrations to the user's hand.

These zones were put together in a larger frame to send vibrations to different parts of the user's hand. The zone near the fingertips needed to transmit high frequencies on the order of 8-15 kHz. This meant the pads needed to cover a large area and be in the front of the device. Then the zone near the base of the fingers and top of the palm needed to transmit middle-range frequencies, which is about 400 Hz to 8 kHz. These were in the middle of the assembly. Finally, the zone near the base of the palm needed to transmit bass frequencies that were between 60 and 400 Hz, which required a smaller pad area.

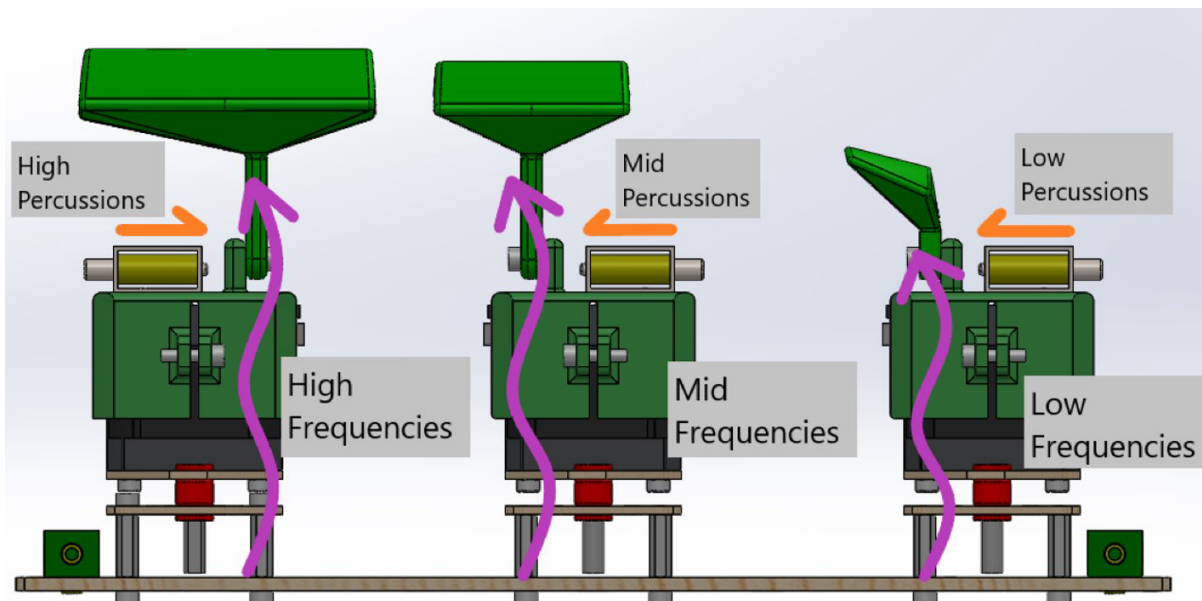


Figure 2: CAD Model of all motor, solenoids and vibration pads setup

While the stepper motors produced tones, the solenoids would emulate percussion in a song. Since percussions also has varying frequencies, different solenoids were specified to certain ranges of frequencies.

All of these components were covered by a large outer case made out of laser cut wooden panels. This allowed the hand to rest on top of the device, avoid touching wires, and stop the zones from touching each other.

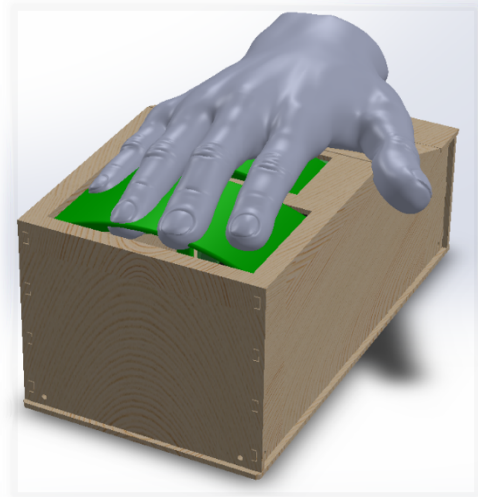


Figure 3: Perspective view of the vibration device within an enclosure

Electronics Implementation

To control the vibration of the stepper motors and when to add the solenoids, a microcontroller was used, which was paired to transistors and H-bridge drivers that interfaced with the actuators. The microcontroller was a Feather M0 Adalogger, which has an onboard micro-SD card reader that enabled the device to read onboard audio files from memory. The adalogger was powered by a 5V USB cable that was connected to a computer during the proof-of-concept phase of the project. The adalogger interfaced with three TB6612 H-bridge circuits which were each powered by a 12V power supply. These H-bridge circuits then connected to the poles of the stepper motors and converted instructions from the adalogger to vibrations for the motors. The adalogger also interfaced with three TIP120 transistors. These transistors were also powered by 12V and the Adafruit interfaced with them over a 1000 Ohm resistor. The solenoids could then be switched on/off when a high or low signal was delivered over the adalogger's digital pins.

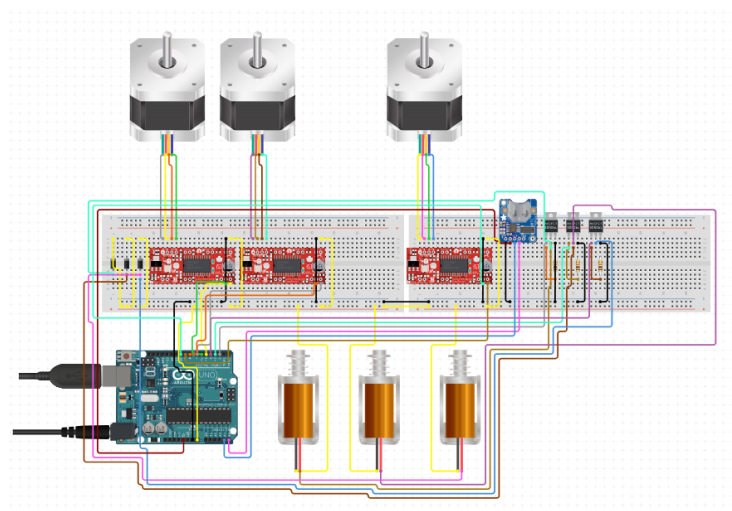


Figure 4: Visual electronics schematic

The physical electronics were soldered to perfboards and loosely placed inside the encasement. Each component was connected using jumper wires that were either soldered together, soldered to the perfboards, or slotted into each other. As a result, the electronics were visually organized, but not structurally organized for typical use.

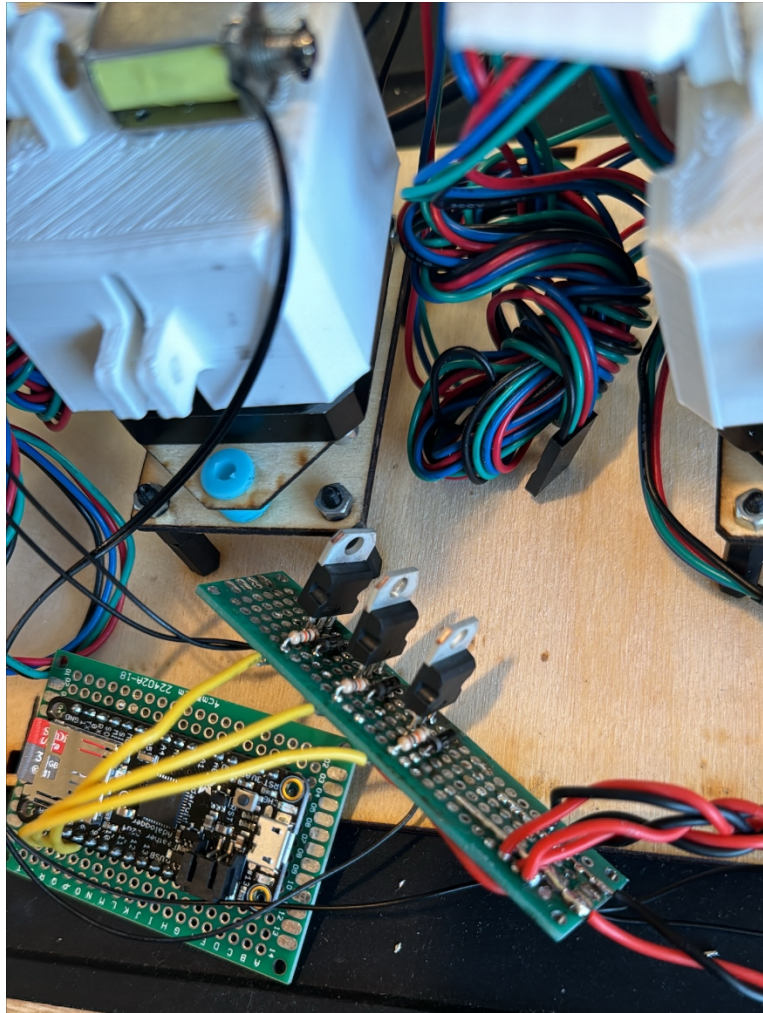


Figure 5: A closeup of the mounting scheme of the electronics

Implementation Algorithm

Audio File Selection and Processing: We selected mp3 audio files of songs that avoid human vocals. The reason behind this choice was to focus on the instrumental aspect of music, as human vocals do not significantly affect the rhythm or flow of the music for our purpose. Moreover, frequency data alone could not effectively represent the presence of human vocals.

Splitting of Music: The music was split into two distinct parts – the Harmonics & the Percussive. The harmonics comprise the melodic part of the music which usually flows over a longer duration of time. It has a tone with a definite pitch and is usually related to instruments like guitars, pianos, and even vocal parts of the song. While percussive on the other have high amplitude over a short period. They represent the rhythmic parts of the music, e.g., the drums or the beats part.

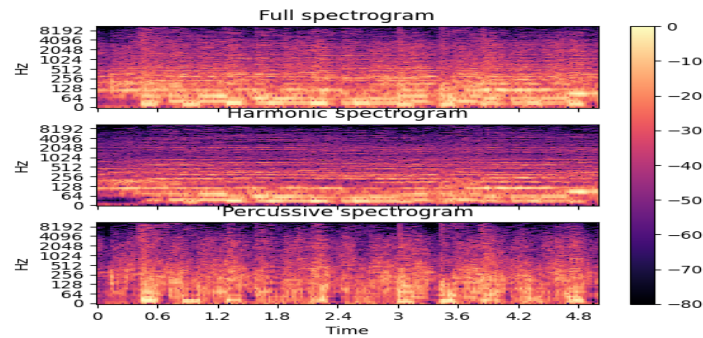


Figure 6: Spectrogram representation of the original, harmonic, and the percussive portion

Percussive for Solenoids: We convert the percussive part to the frequency domain using Fast Fourier Transform (FFT), then it was divided into 3 frequency ranges, namely low, mid, and high. Then each band was converted back into the time domain using Inverse Short Time Fourier Transform (ISTFT)

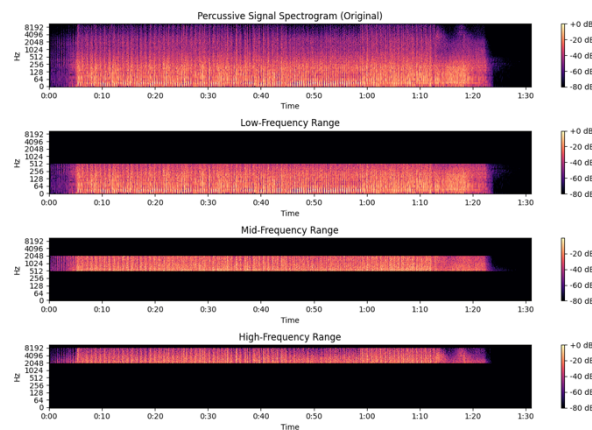


Figure 7: Spectrogram representation after the converting

Noise removal: The signals obtained would have different amplitudes, hence a noise-removing threshold based on 70% of the original amplitude was implemented, to get only the major contributing beat in that frequency range. A column binary array with 1 representing the existence of a beat on that time stamp, and 0 otherwise was initialized. Then this array is assigned 1 each for the three different frequency ranges and was stored as a CSV file. Now this file would be read inside the microcontroller and a logic High for 1 and Low for 0 would be applied to the solenoid to give the feeling of a beat.

Conversion to Frequency Data: We used the Librosa library in Python 3 [4], and audio files were converted into frequency data. Librosa library contains an import feature extraction process, the one we mainly focused on is Chroma features, which represent the energy distribution across 12 different pitch classes and are useful in analyzing music where pitch and harmony are important. Then the Short-Time Fourier Transform (STFT) [5] is applied to the signal to obtain the frequency spectrum. Frequencies are mapped to the 12 different pitch classes in the chromatic scale (e.g., C, C#, D, D#, etc.). The energy in each pitch class is summed up, resulting in a 12-element chroma vector for each time frame.

Mel Frequency Cepstral Coefficients (MFCC): We planned another approach using MFCC. MFCC is excellent for capturing the timbral aspects of sound but can be less effective for rhythmic or pitch-based analysis. MFCCs typically produce a high-dimensional feature space. Each frame (a short segment of audio) is represented by a vector of MFCC values (commonly around 13 to 20 coefficients)[6]. In our case, a large matrix (2000 x 2250) suggests a high number of frames with many coefficients per frame, which is excessive for controlling just three motors. At the same time, a high prevalence of zeroes might indicate periods of silence or low energy in the audio. This can lead to over-segmentation, where the rhythm continuity is lost, as MFCCs are more sensitive to timbral changes than to rhythm or pitch.

Time	Dominant Frequency	categories	low	mid	high
0.1	300	low	1	0	0

Data processing: In the data processing approach for controlling stepper motors with audio input, we convert audio files into a structured CSV format with a focus on key attributes. As shown in the table, each row of the CSV represents a 0.1-second timeframe of the audio, aligning with the average human touch reaction time of 0.15 seconds, as researched by Backyard Brains [7]. This timeframe ensures that changes in the music are captured at a rate perceivable by human listeners. Additionally, each motor is programmed with a 100 ms delay to detect and react to changes in the dominant frequency, simplifying the audio data. The dominant frequency, calculated as an average of grouped frequencies from the song, is used to categorize the audio into high, mid, and low intervals. These intervals then correspond to the specific actions of the three motors, providing a coherent and manageable way to translate audio characteristics into motor movements, creating a system that responds to music in a manner akin to human auditory perception. Upon completion of the data processing, the processed audio data is stored on an SD card, which serves as an intermediary for transferring the information from the PC to the microcontroller. This approach utilizes the SD card as a bridge, effectively facilitating the conveyance of the refined audio data to the microcontroller for subsequent actions.

Signal processing: In the signal processing segment of our project, we employed standard header files, specifically `stepper.h` [8] and `sd.h`, along with predefined templates to manage the stepper motors and solenoids. This setup encompassed initialization procedures, parameter definitions, and essential configurations in the setup function. During setup, we executed a critical operation of reading the SD card, specifically processing a CSV file line by line. This method allowed us to extract each row's values and store them in separate variables for future use in the loop function. Within the loop function, we then processed each row of data, allocating the parsed values to their respective fields. Based on these values, we implemented conditional checks to determine the appropriate responses for each stepper motor and solenoid, ensuring that their actions were in sync with the processed audio data. This methodology enabled a dynamic and responsive system, translating the analysed audio signals into precise mechanical movements.

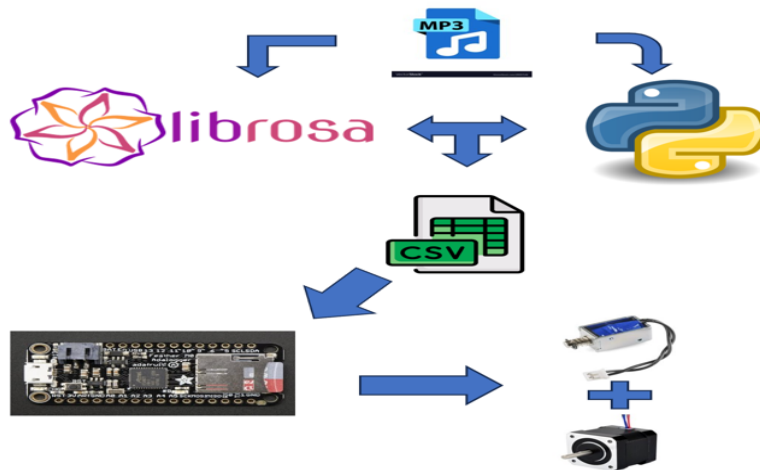


Figure 8: Sketch of execution methodology

Operation logic: In the revised operational logic of our project, the microcontroller processes data to represent low, mid, and high-frequency intervals instead of generating random numbers. This adjustment more accurately mirrors the diverse range of frequencies found in music. For each frequency interval (low, mid, high), we have a designated stepper motor: the first motor corresponds to low frequencies, the second to mid frequencies, and the third to high frequencies. The movements of these motors are calculated based on their specific speeds and steps per revolution, ensuring that each motor's motion is synchronized with the frequency it represents. Additionally, the control of the solenoids is directly derived from the CSV file, which contains three dedicated columns representing low, mid, and high frequencies for solenoid activation. In these columns, a value of 0 or 1 is used to dictate the solenoid's movement: 0 indicates a backward motion, while 1 indicates a forward motion. This setup is particularly effective in simulating the impact of drumbeats, where a forward motion (1) can emulate the sensation of a drum being hit. The system loops through the CSV data, activating the respective stepper motor and solenoids based on these values. This method creates a dynamic and rhythmic pattern of mechanical movements and vibrations, thereby translating the essence of the music into a tactile experience. The delays between activations are adjustable, allowing the system to be fine-tuned for a realistic and immersive interpretation of music through touch.

RESULTS

In our recent demonstration, we successfully showcased the audio-to-touch device by playing a song with and without solenoids, effectively highlighting the variation in output. The device's components functioned as intended, and images showcasing the prototype are provided for reference. Despite the successful performance, there are several areas identified for improvement:

Aesthetic Design: The overall appearance of the product offers room for enhancement to achieve a more aesthetically pleasing look. A more appealing design could significantly improve the user appeal.

Precision in Construction: We encountered some inaccuracies in the sizing and length measurements of the product's parts, leading to imperfect assembly. As a result, certain components did not align or fit evenly, impacting the effectiveness of the vibration feedback. Attention to detail in measurements and assembly is crucial for optimal performance.

Material Quality: Due to budget constraints, the materials used were basic, posing a risk of damage during transportation and usage. Investing in higher-quality materials could enhance the product's durability and tactile experience. Exploring alternative materials could offer a better balance between cost and quality.

Operational Parameters: The current operational logic of the device is somewhat limited, having been tested for short durations only. This raises concerns about potential bugs and risks during extended operation. To cater to customers desiring a continuous playlist experience, additional functionality and control logic are necessary. This enhancement would allow for seamless transitions between songs and a more robust user experience.

Wiring and Internal Layout: The wiring process presented significant challenges, primarily due to miscalculations in the spatial arrangement and the rigidity of the materials used. The current wiring layout appears disorganized and could benefit from a more structured design. Improving the internal layout would not only enhance the device's functionality but also its overall durability and user safety.

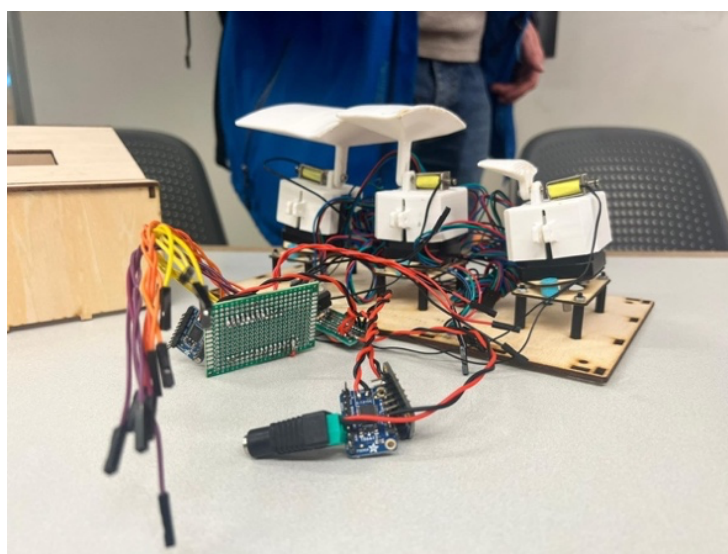


Figure 9: Device without top cover to display the electronics and hardware.



Figure 10: The top view of the device with housing and holes laser cut in it to expose the transmitting pads

CONCLUSION

We have successfully demonstrated a proof of concept by building and successfully executing the principle. In summary, while the prototype demonstrates the core functionality successfully, refinements in design, materials, operational logic, and internal layout are essential for the next development phase. Our prototype testing revealed promising results in its core functionality, providing a novel way for the hearing-impaired to experience music. However, the project also highlighted key areas requiring further development, including the aesthetic design of the product, the precision of its construction, the quality of materials used, and the refinement of operational parameters and internal layout. These improvements are essential for optimizing the user experience, ensuring durability, and enhancing the practical application of the device. In future phases of this project, we plan to address these challenges. Our goal is to refine the design and functionality, with a focus on user accessibility and comfort, to better cater to the needs of our target audience. This endeavour not only represents a technological innovation but also a commitment to inclusive design, emphasizing the role of engineering in creating solutions that are accessible and beneficial to all segments of society. These improvements aim to elevate the user experience, ensure long-term reliability, and enhance the product's viability.

The video displaying the functionality has been linked through a YouTube video: https://www.youtube.com/watch?v=Ov1A_Oo0Y4c.

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